From Raychaudhuri's Equation to Neutron-Star Collapse

A worked study: geodesic focusing, causal equations of state, hydrostatic equilibrium, and what changes when collapse is genuinely dynamical

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1. Theoretical inputs

1.1 Raychaudhuri's equation

Everything in this study traces back to one equation, governing the expansion scalar θ of a congruence of worldlines with tangent u^a:

$$\dot{\theta} = -\frac{\theta^2}{3} - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}u^a u^b + \nabla_a a^a$$

where σ_{ab} is the shear, ω_{ab} the vorticity, $R_{ab}u^a u^b$ the Ricci curvature sourced by matter, and $a^a = u^b \nabla_b u^a$ the acceleration of the congruence — nonzero only when the worldlines are not geodesic (e.g. fluid elements pushed by a pressure gradient). For a perfect fluid:

$$R_{ab}u^a u^b = 4\pi G(\rho + 3p), \qquad a_a = -\frac{\nabla_a p}{\rho + p}$$

1.2 Specializations explored, in order

Constant-source phase line:

$$\dot{\theta} = -\frac{\theta^2}{3} - \Phi$$

for constant Φ — the textbook demonstration that $\Phi>0$ (strong energy condition) forces focusing to $\theta \to -\infty$ in finite proper time, with no equilibrium.

Shear and vorticity:

$$\dot{\theta} = -\frac{\theta^2}{3} - (\Phi + \sigma^2 - \omega^2)$$

Shear always accelerates focusing; vorticity is the only term that can halt it.

FRW-coupled system: $\theta=3H$, coupled to $\rho\blacksquare=-\theta(\rho+p)$ with $p=w\rho$, producing a genuine 2D dynamical system whose fixed points (for $w=-1$) reproduce the de Sitter attractor.

1.3 A causal, neutron-star-consistent equation of state

To model a collapsing cloud whose matter stiffens at high density without violating causality:

$$p(\rho) = \rho \frac{x}{1+x}, \qquad x = \left(\frac{\rho}{\rho_c}\right)^4$$

dust-like ($p\approx 0$) for $\rho \ll \rho_c$, approaching the causal limit $p \to \rho$ (sound speed $\to c$) for $\rho \gg \rho_c$ — the same logic behind the Rhoades–Ruffini absolute bound on neutron-star mass.

1.4 Hydrostatic structure: the TOV equations

For a genuine static equilibrium ($G=c=1$):

$$\frac{dm}{dr} = 4\pi r^2 \rho, \qquad \frac{dp}{dr} = -\frac{(\rho+p)(m+4\pi r^3 p)}{r(r-2m)}$$

Solved by shooting outward from many central densities, giving a mass–central-density curve with a genuine maximum: the **turning-point theorem** (Bardeen–Thorne–Meltzer) says stability requires $dM/d\rho_{center}>0$, so the first maximum of $M(\rho_{center})$ is the true M_{max}.

1.5 An EoS-only stability proxy: the adiabatic index

$$\Gamma \equiv \frac{d\ln p}{d\ln \rho} = \frac{x+5}{1+x}$$

for our EoS, crossing the classical marginal-stability value $\Gamma=4/3$ at $\rho\approx 1.82\,\rho_c$ — independent of mass or radius, a purely local diagnostic that agreed with the TOV result's order of magnitude.

1.6 The dynamical extension: putting $\nabla_a a^a$ back

A homogeneous cloud has zero pressure *gradient*, so $a^a\equiv 0$ identically — which is why purely homogeneous collapse, however stiff the EoS, always focuses to a singularity ($\rho+3p\geq 0$ is never violated). To restore a genuine acceleration term without solving the full PDE, we used a top-hat closure with characteristic radius $R(\tau)$, $R\blacksquare=-\theta R/3$:

$$\nabla_a a^a \approx \frac{k\,p}{R^2(\rho+p)}$$

giving the reduced 3-variable system used for essentially all of the numerical work below:

$$\dot{\theta} = -\frac{\theta^2}{3} - (\rho + 3p) + \frac{kp}{R^2(\rho+p)}, \quad \dot{\rho} = -\theta(\rho+p), \quad \dot{R} = \frac{\theta R}{3}$$

1.7 What fully determines a trajectory

Because this is an autonomous ODE system, a trajectory (and everything derived from it) is completely fixed by: the **equation of state** (ρ_c, k), and the **initial data** (ρ_0, R_0, θ_0) — not just the initial mass M_0. A numerical check (§3.7) showed that fixing M_0 and θ_0 while varying R_0 alone changes the outcome.

2. Computational approach

Two complementary tools were used throughout: **interactive exploration** (in-chat widgets) for the early conceptual stages — the constant-Φ phase line, the shear/vorticity extension, the FRW phase plane, the homogeneous collapsing-cloud demonstration, the first TOV solve, and the first (θ,ρ) phase portrait with the support term — and **verified numerical computation** (Python: NumPy + Matplotlib, run and inspected before presenting) for everything quantitative from the mass–velocity stability diagram onward. This is also where an actual bug — a sign error feeding expansion rates instead of collapse rates into the first stability-diagram attempt — was caught by checking the output against hand-computed values before presenting it.

Core numerical methods: explicit Euler integration of the 3-variable ODE system (step sizes 10^{-3}–10^{-4}); bisection search on initial density to locate critical/separatrix masses; a shooting method with an inner bisection (to invert $p(\rho)$, which has no closed form) for the TOV solves.

3. Results

3.1 The modified stability diagram — the central result

What looked like one number, M_{max}, from hydrostatic (TOV) theory is really the zero-velocity limit of a full boundary curve in the plane of initial mass and initial infall rate.

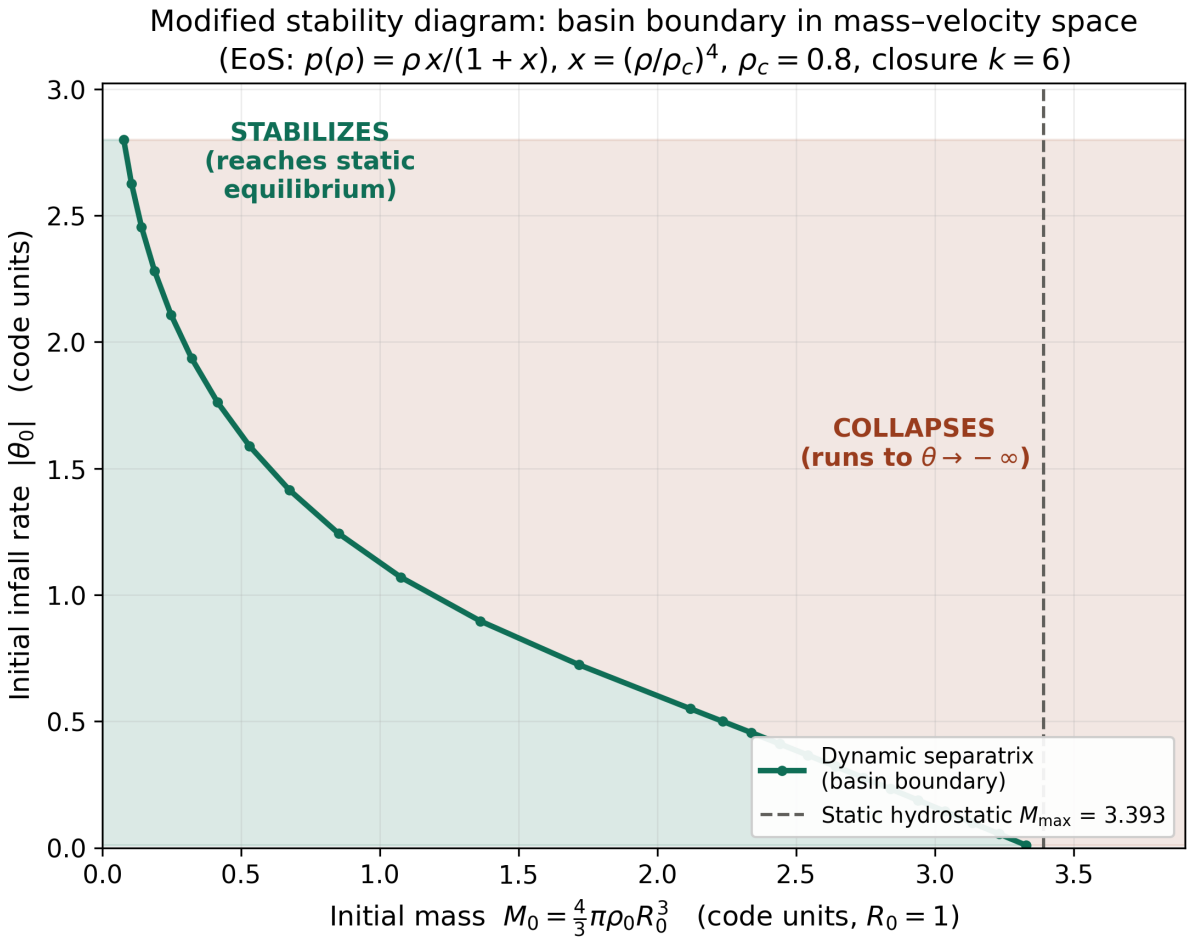


Fig. 1 — Separatrix between clouds that dynamically stabilize (green) and clouds that collapse (red) in the plane of initial mass M0 and initial infall rate |θ0|. The static hydrostatic M_max (dashed) is only the θ0→0 limit of this curve; by |θ0|≈1, the surviving mass is already under a third of the static bound.

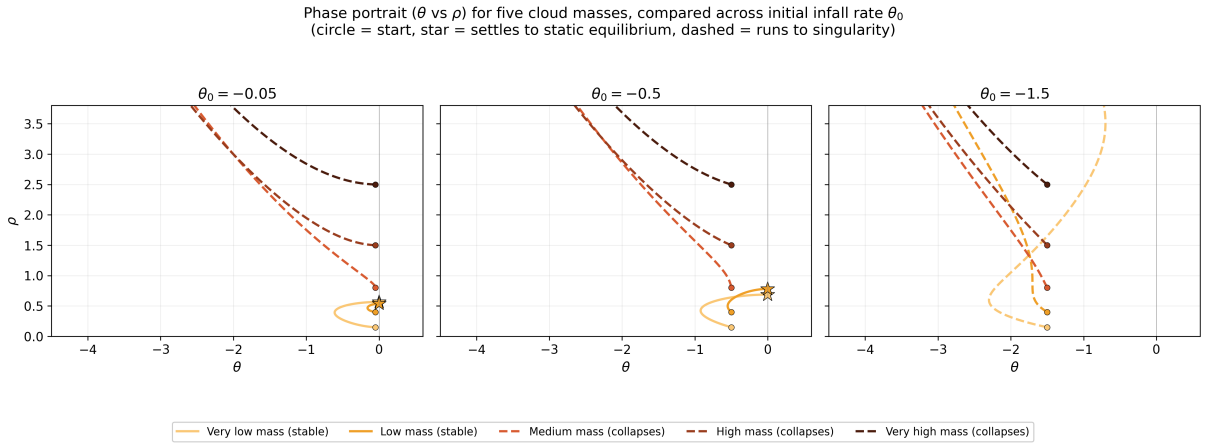


Fig. 2 — The same conclusion as trajectories: five cloud masses run at three infall rates. At low |θ0| the lightest two masses settle to equilibrium (star markers); by θ0=-1.5 even the lightest mass is swept into collapse (dashed).

3.2 Trajectories in θ■–θ and θ–τ

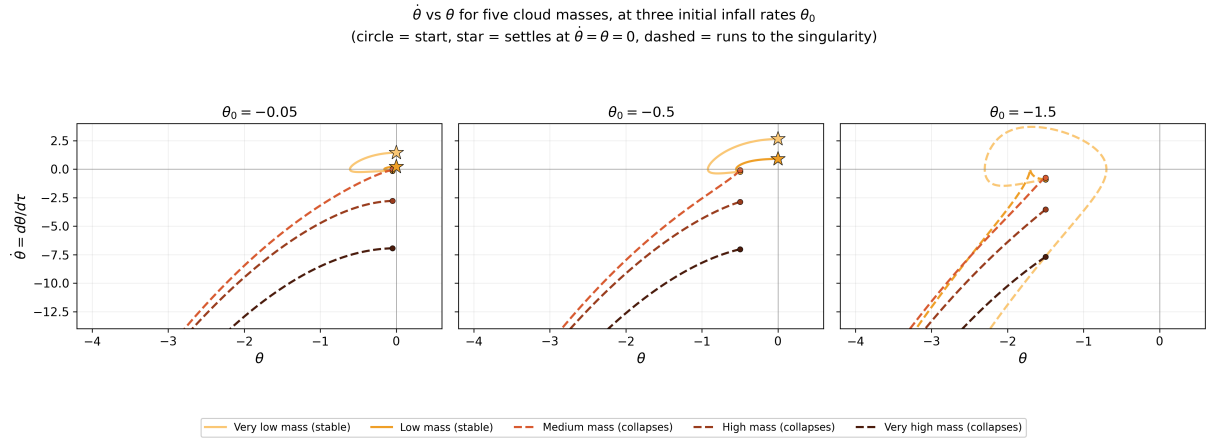


Fig. 3 — theta-dot vs theta along real trajectories: stabilizing masses spiral into the origin (star, where theta-dot = theta = 0 simultaneously); collapsing masses plunge downward without ever recrossing theta-dot = 0.

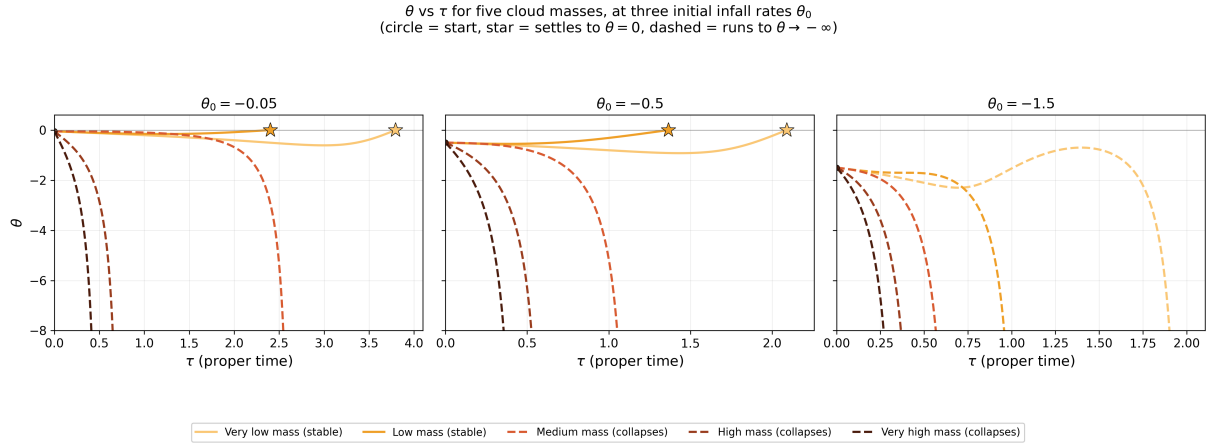


Fig. 4 — theta vs proper time tau for the same runs. Surviving trajectories visibly overshoot upward before settling — direct evidence of inertial “ringing” from the kinetic term.

3.3 Decomposing θ■ into its physical pieces

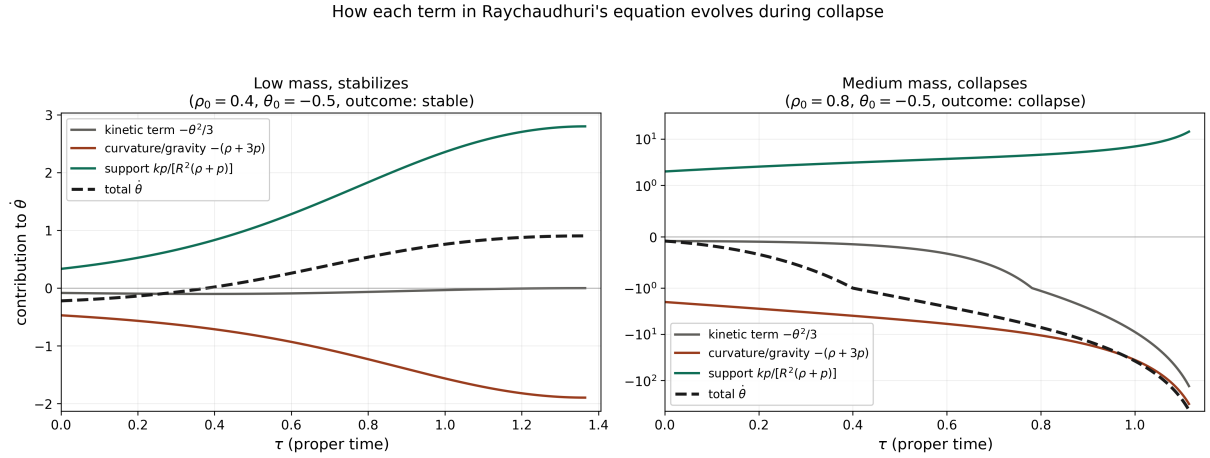


Fig. 5 — Left: support (teal) climbs above gravity (red) around tau ~ 0.3-0.4 and the total (dashed) turns around. Right (log scale): support climbs by over an order of magnitude but never gets ahead of gravity, and the kinetic term joins in near the end.

3.4 Do support and gravity always mirror each other?

Testing across masses from deep-stable to deep-collapse showed the mirroring seen earlier does not hold generically — it is a signature specifically of trajectories near the critical mass:

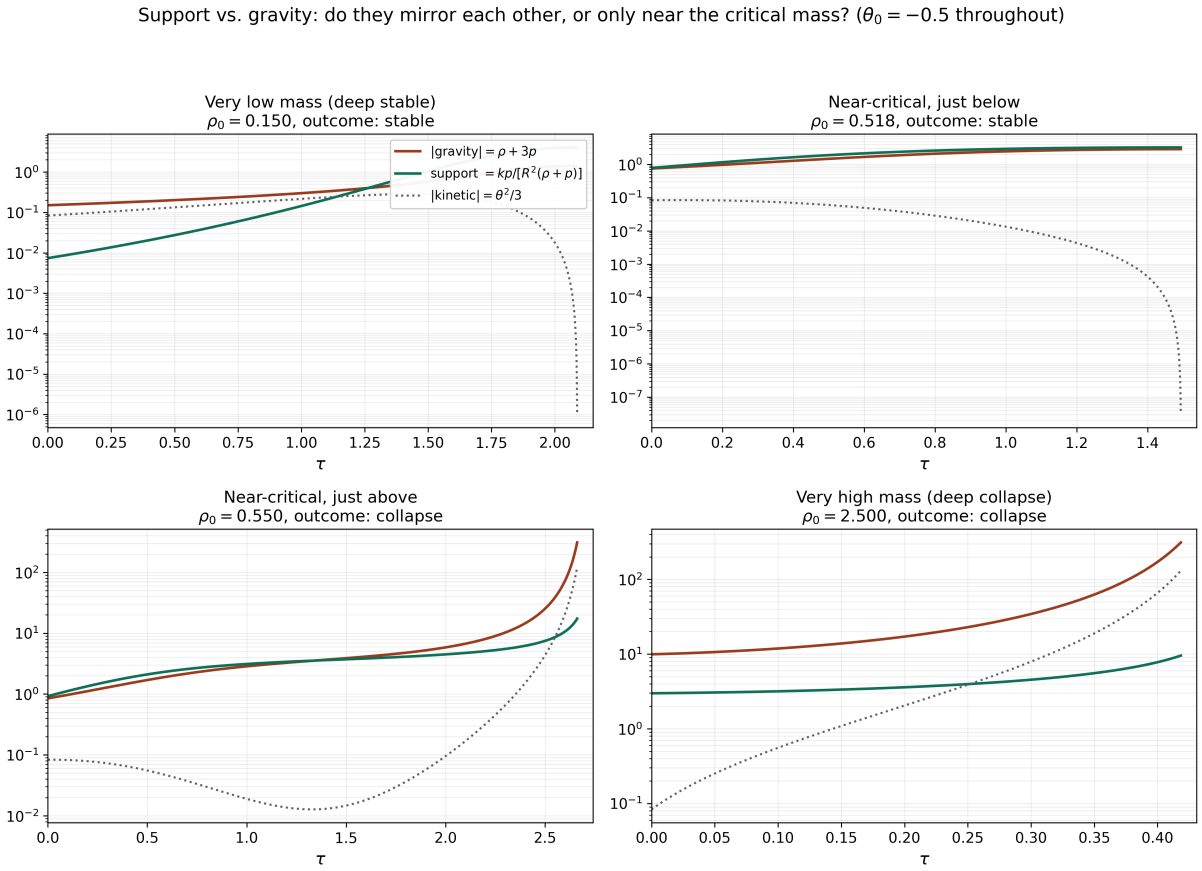


Fig. 6 — |gravity| and support, log scale, across four masses at fixed theta0=-0.5. Deep-stable and deep-collapse cases separate immediately; only the near-critical cases show extended close tracking.

Fig. 7 — Same four cases with signs intact (symlog scale). Support and gravity never actually cross; it is the kinetic term's trend that decides the outcome in the near-critical cases.

3.5 Why support and gravity share the same shape

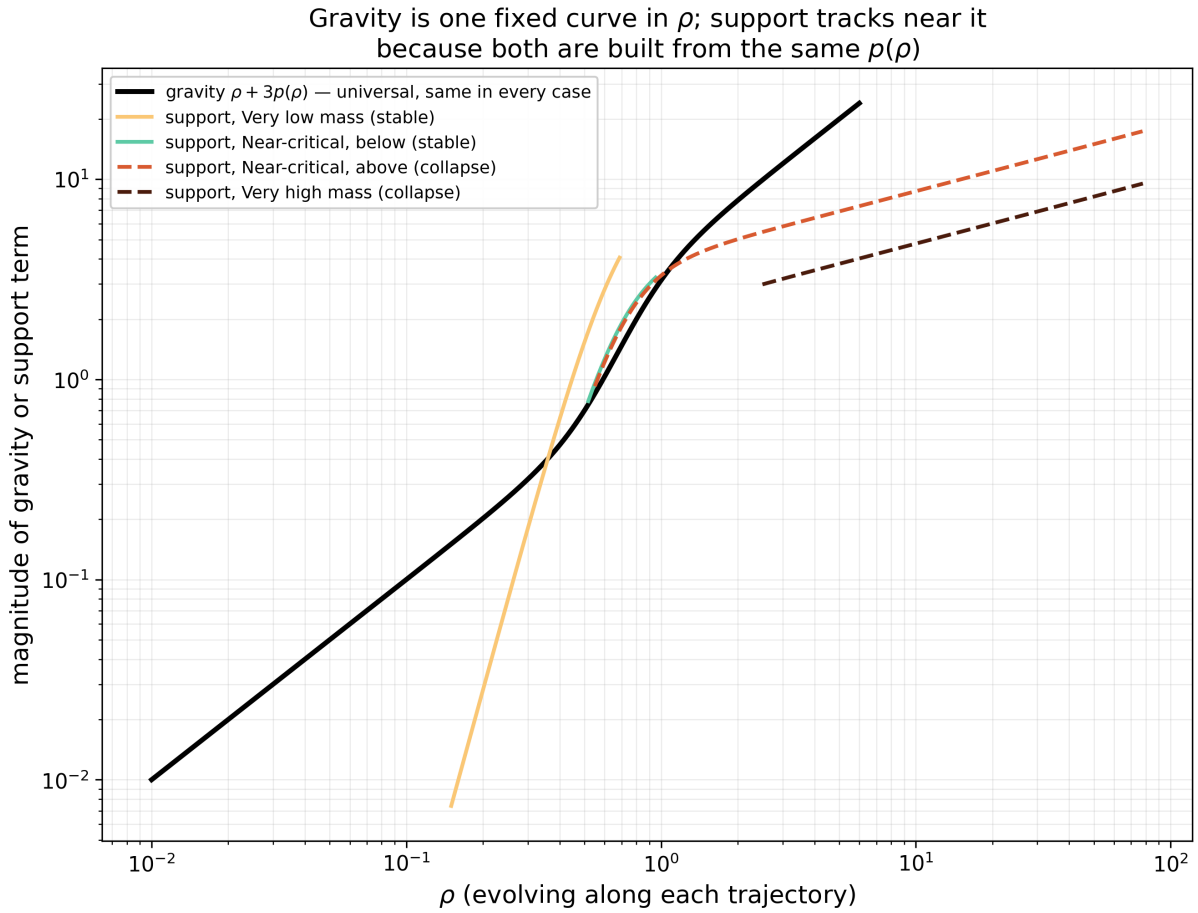


Fig. 8 — Gravity plotted against rho (not tau) is a single universal curve, identical for every mass. Every support curve threads close to that same curve, because it is built from the same p(rho), just rescaled by 1/R^2.

3.6 How much work is R actually doing?

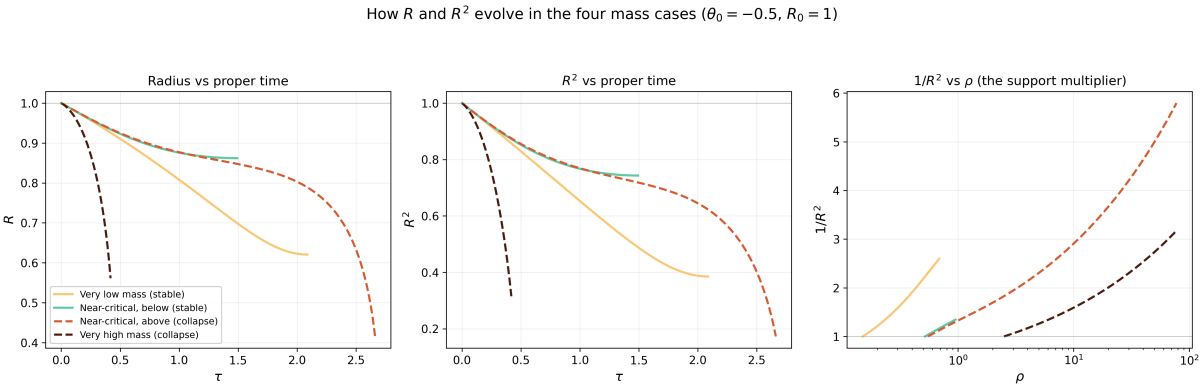


Fig. 9 — R and R^2 across the four cases. R never moves by more than roughly a factor of 2.4 in any run, a mild correction compared to the many-decade swing of rho and p(rho).

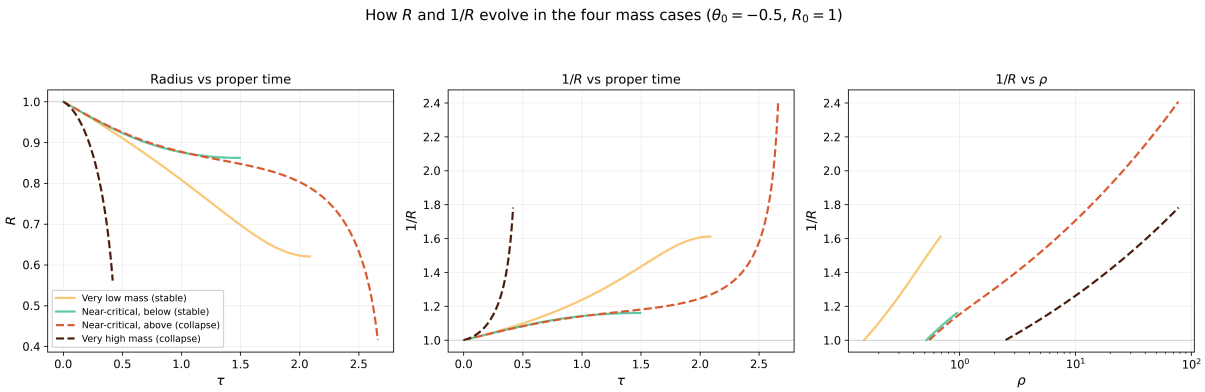


Fig. 10 — R and 1/R directly. Monotonic in every case, as expected since theta < 0 throughout the plotted range.

3.7 Mass alone is not enough

Holding M0=2.0 and theta0=-0.5 fixed while varying R0 (adjusting rho0 to compensate):

R0	rho0	Compactness M0/R0	Outcome
0.6	2.21	3.33	collapses
0.8	0.93	2.50	collapses
1.0	**0.48**	**2.00**	**STABLE**
1.2	0.28	1.67	collapses
1.5	0.14	1.33	collapses
2.0	0.06	1.00	collapses

Same mass, same initial velocity, six different outcomes — and non-monotonically in R0. Initial compactness is an independent input, not something M0 alone fixes.

4. Summary & key conclusions

• **Geodesic focusing is unconditional under the strong energy condition.** A homogeneous cloud, however stiff its EoS (up to the causal limit), always collapses — and denser clouds collapse *faster*, because pressure itself gravitates in GR (ρ+3p sources curvature).

• **Real stellar support needs $\nabla_a a^a \neq 0$** — a genuine pressure gradient, which requires inhomogeneity. This is exactly what the homogeneous model structurally lacks, and exactly what the TOV equations supply.

• **The static maximum mass is real but incomplete.** The turning-point theorem gives a rigorous M_{max} for equilibria at rest; the EoS-only adiabatic-index criterion (Γ=4/3) gives a quick, purely local estimate of the same scale.

• **Dynamics matters, often a great deal.** A cloud already falling inward carries kinetic focusing power ($-\theta^2/3$) that has nothing to do with its EoS. The static M_{max} is only the $\theta 0 \to 0$ edge of a full 2D separatrix in (M0, |θ0|).

• **Support and gravity share a common origin** (the same p(ρ)) so they trend together in shape; what decides stability is their near-cancellation, and the sign of the kinetic term's drift once that cancellation is tested.

• **Mass is necessary but not sufficient input.** The full determining data is the EoS plus (ρ0, R0, θ0) — initial compactness carries independent information that total mass alone discards.

5. Caveats & scope

• The equation of state $p(\rho)=\rho x/(1+x)$ is an illustrative causal-stiffening toy, not a fitted nuclear EoS; quoted masses and densities are in arbitrary code units, not solar masses.

• The acceleration-divergence closure $kp/[R^2(\rho+p)]$ is a crude, order-unity top-hat estimate for $\nabla_a a^a$, not a rigorous derivation. Specific critical masses from this closure should be read qualitatively, not quantitatively.

• Integration used explicit Euler with small fixed steps; adequate here, but not a substitute for an adaptive relativistic hydrodynamics code for precision work.

• Real astrophysical numbers (neutron star M_max ~ 2-3 solar masses) were quoted only as literature context, not reproduced by this toy model.

6. Reproducibility: core code

The physics kernel used for every quantitative result in Sections 3.1–3.7 (Python):

```
rc_ = 0.8   # nuclear stiffening scale, rho_c
k_  = 6.0   # support-closure coefficient

def eos(rho):
    x = (rho/rc_)**4
    return rho*x/(1+x)          # p(rho): dust at low rho, causal (p->rho) at high rho
```

```
def rhs(theta, rho, R):
    p = eos(rho)
    kinetic = -theta**2/3
    gravity = -(rho + 3*p)
    support = k_*p/(R**2*(rho+p))
    thetadot = kinetic + gravity + support
    rhodot   = -theta*(rho+p)
    Rdot     = theta*R/3
    return thetadot, rhodot, Rdot

def integrate(rho0, theta0, R0=1.0, dt=0.001, tmax=16.0):
    theta, rho, R, tau = theta0, rho0, R0, 0.0
    while tau < tmax and rho < 80 and R > 1e-3 and theta > -30:
        thd, rhod, Rd = rhs(theta, rho, R)
        theta += dt*thd; rho += dt*rhod; R += dt*Rd; tau += dt
        if theta > -1e-4 and tau > 0.25:
            return 'stable', tau
    return 'collapse', tau

def critical_rho0(theta0, lo=0.003, hi=6.0, iters=26):
    for _ in range(iters):
        mid = (lo+hi)/2
        outcome, _ = integrate(mid, theta0)
        lo, hi = (mid, hi) if outcome == 'stable' else (lo, mid)
    return (lo+hi)/2
```

The independent hydrostatic (TOV) check used a standard shooting method: integrate dm/dr, dp/dr outward from r→0 with central density, inverting p(ρ) at each step by bisection, until p→0 defines the stellar surface; repeating across central densities traces out $M(\rho_{center})$ and its maximum.